

Multi-dimensional Runge Kutta Stability

CPS 406

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Introduction

- **Context:** Many engineering problems are modeled using ordinary differential equations. When a problem has more than one changing quantity, we write it in the form of system of differential equations.
- **Problem Statement:** If the exact solution of a system should decay smoothly, a numerical method can become unstable if the step size is too large.
- **Objective:** This presentation shows how step size affects stability for a simple two-dimensional Ordinary Differential Equation system using Euler's method and the fourth-order Runge-Kutta method.

Core Concept

Stability asks whether the numerical solution behaves like the exact solution. If the exact solution decays, the numerical solution should also decay instead of growing.

Two dimensional ODE System

For $y_1' = -2y_1$, (initial : $y_1(0) = 1$):

$$\frac{dy_1}{dt} = -2y_1$$

$$\frac{1}{y_1} dy_1 = -2dt$$

$$\int \frac{1}{y_1} dy_1 = \int -2dt$$

$$\ln |y_1| = -2t + C$$

$$y_1(t) = Ce^{-2t}$$

$$y_1(0) = 1 \implies 1 = Ce^0 \implies C = 1$$

$$\mathbf{y_1(t) = e^{-2t}}$$

For $y_2' = -5y_2$, (initial : $y_2(0) = 1$):

$$\frac{dy_2}{dt} = -5y_2$$

$$\frac{1}{y_2} dy_2 = -5dt$$

$$\int \frac{1}{y_2} dy_2 = \int -5dt$$

$$\ln |y_2| = -5t + C$$

$$y_2(t) = Ce^{-5t}$$

$$y_2(0) = 1 \implies 1 = Ce^0 \implies C = 1$$

$$\mathbf{y_2(t) = e^{-5t}}$$

Expected Behavior

Since both exact solutions decay to zero ($e^{-2t}, e^{-5t} \rightarrow 0$), any stable numerical method must also produce decaying values.

ODE Methods as Numerical Integration

To show you that: ODE solving $\approx$ approximating an integral

- 1 **Start with the ODE:** $y'(t) = f(t, y)$
- 2 **Integrate in the time step** $[t_n, t_{n+1}]$: Let $h(\text{step size}) = t_{n+1} - t_n$. Then:

$$\int_{t_n}^{t_{n+1}} y'(t) dt = \int_{t_n}^{t_{n+1}} f(t, y(t)) dt$$

- 3 **Apply the Fundamental Theorem of Calculus:** The left-hand side simplifies directly:

$$y(t_{n+1}) - y(t_n) = \int_{t_n}^{t_{n+1}} f(t, y(t)) dt$$

The Fundamental Integral Equation

After adding $y(t_n)$ on both sides, we get the basis for all one-step numerical methods:

$$y(t_{n+1}) = y(t_n) + \int_{t_n}^{t_{n+1}} f(t, y(t)) dt$$

Euler's Method for the 2D System

Intuition: Euler's method approximates the integral by assuming that the slope $f(t, y)$ forms a constant rectangle of height $f(t_n, y_n)$ over width h . Since area of the rectangle $\Rightarrow w \times h \Rightarrow hf(t_n, y_n)$

Step 1: Vector Form

Define the discrete state and slope vectors from our original ODE system:

$$\mathbf{y}_n = \begin{bmatrix} y_{1,n} \\ y_{2,n} \end{bmatrix}, \quad \mathbf{f}(t_n, \mathbf{y}_n) = \begin{bmatrix} -2y_{1,n} \\ -5y_{2,n} \end{bmatrix}$$

Step 2: Substitution

Substituting these vectors into our formula $\mathbf{y}_{n+1} = \mathbf{y}_n + hf(t_n, \mathbf{y}_n)$, we get:

$$\begin{bmatrix} y_{1,n+1} \\ y_{2,n+1} \end{bmatrix} = \begin{bmatrix} y_{1,n} \\ y_{2,n} \end{bmatrix} + h \begin{bmatrix} -2y_{1,n} \\ -5y_{2,n} \end{bmatrix}$$

Step 3: Component-wise Addition

Distributing step size h and then adding component by component into one system:

$$\begin{bmatrix} y_{1,n+1} \\ y_{2,n+1} \end{bmatrix} = \begin{bmatrix} y_{1,n} - 2hy_{1,n} \\ y_{2,n} - 5hy_{2,n} \end{bmatrix}$$

Step 4: Factor

Further factoring will yield two formulas. The multipliers will be later important for stability:

Final Update Formulas

$$y_{1,n+1} = (1 - 2h)y_{1,n}$$

$$y_{2,n+1} = (1 - 5h)y_{2,n}$$

Euler Stability Analysis: Component 1

The Stability Idea:

From our Euler formula update, we found that each step multiplies the old value by a specific factor:

$$y_{1,n+1} = (1 - 2h)y_{1,n}, \quad y_{2,n+1} = (1 - 5h)y_{2,n}$$

For the exact solutions (e^{-2t} and e^{-5t}), we saw that both components decay to zero. So, for the numerical method to decay, the multipliers must have an absolute value of less than 1:

$$|1 - 2h| < 1 \quad \text{and} \quad |1 - 5h| < 1$$

First Condition (y_1)

$$|1 - 2h| < 1$$

$$-1 < 1 - 2h < 1 \quad (\text{Expanding})$$

$$-2 < -2h < 0 \quad (\text{Subtracting 1 on both sides})$$

$$1 > h > 0 \quad (\text{Dividing by } -2, \text{ and flipping inequality signs})$$

Rewriting in increasing order, the first component is stable when:

$$\mathbf{0 < h < 1}$$

Euler Stability Analysis: Component 2

Second Condition (y_2)

$$|1 - 5h| < 1$$

$$-1 < 1 - 5h < 1 \quad (\text{Expanding})$$

$$-2 < -5h < 0 \quad (\text{Subtracting 1 on both sides})$$

$$\frac{2}{5} > h > 0 \quad (\text{Dividing by } -5, \text{ and flipping inequality signs})$$

Rewriting in increasing order, the second component is stable when:

$$0 < h < 0.4$$

Combine Both Conditions:

The full system is stable only if *both* components are stable simultaneously.

- First component requires: $0 < h < 1$ and Second component requires: $0 < h < 0.4$

The stricter condition dominates. So, Euler's method is stable for the whole 2D system only when:

$$0 < h < 0.4$$

Euler Example: Stable Step Size

So, the stability condition: $0 < h < 0.4$. Let's test a step size, $h = 0.1$.

1. Calculate Multipliers

Substituting $h = 0.1$ into our update formulas:

$$\begin{aligned}y_{1,n+1} &= (1 - 2(0.1))y_{1,n} \\ &= (1 - 0.2)y_{1,n}\end{aligned}$$

$$\mathbf{y}_{1,n+1} = \mathbf{0.8y}_{1,n}$$

$$\begin{aligned}y_{2,n+1} &= (1 - 5(0.1))y_{2,n} \\ &= (1 - 0.5)y_{2,n}\end{aligned}$$

$$\mathbf{y}_{2,n+1} = \mathbf{0.5y}_{2,n}$$

2. Iterate the System

Start with our initial condition $\mathbf{y}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$:

$$\mathbf{y}_1 = \begin{bmatrix} 0.8(1) \\ 0.5(1) \end{bmatrix} = \begin{bmatrix} 0.8 \\ 0.5 \end{bmatrix}$$

$$\mathbf{y}_2 = \begin{bmatrix} 0.8(0.8) \\ 0.5(0.5) \end{bmatrix} = \begin{bmatrix} 0.64 \\ 0.25 \end{bmatrix}$$

$$\mathbf{y}_3 = \begin{bmatrix} 0.8(0.64) \\ 0.5(0.25) \end{bmatrix} = \begin{bmatrix} 0.512 \\ 0.125 \end{bmatrix}$$

Result

Both components strictly decay toward zero. The numerical method is **stable**.

Euler Example: Unstable Step Size

Now let's test $h = 0.5$.

1. Calculate Multipliers

Substitute $h = 0.5$ into our update formulas:

$$\begin{aligned}y_{1,n+1} &= (1 - 2(0.5))y_{1,n} \\ &= (1 - 1)y_{1,n}\end{aligned}$$

$$\mathbf{y}_{1,n+1} = \mathbf{0} \cdot \mathbf{y}_{1,n} = \mathbf{0}$$

$$\begin{aligned}y_{2,n+1} &= (1 - 5(0.5))y_{2,n} \\ &= (1 - 2.5)y_{2,n}\end{aligned}$$

$$\mathbf{y}_{2,n+1} = -1.5\mathbf{y}_{2,n}$$

2. Iterate the System

Start with initial condition $\mathbf{y}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$:

$$\mathbf{y}_1 = \begin{bmatrix} 0(1) \\ -1.5(1) \end{bmatrix} = \begin{bmatrix} 0 \\ -1.5 \end{bmatrix}$$

$$\mathbf{y}_2 = \begin{bmatrix} 0(0) \\ -1.5(-1.5) \end{bmatrix} = \begin{bmatrix} 0 \\ 2.25 \end{bmatrix}$$

$$\mathbf{y}_3 = \begin{bmatrix} 0(0) \\ -1.5(2.25) \end{bmatrix} = \begin{bmatrix} 0 \\ -3.375 \end{bmatrix}$$

Result

The second component's magnitude grows ($1 \rightarrow -1.5 \rightarrow 2.25 \rightarrow -3.375 \dots$). The numerical method is **unstable**.

RK4 Derivation: Computing $\mathbf{k}_1$ and $\mathbf{k}_2$

System Setup: $\mathbf{f}(t, \mathbf{y}) = \begin{bmatrix} -2y_1 \\ -5y_2 \end{bmatrix}$, Initial: $\mathbf{y}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, Step size: $h = 0.1$

Step 1: Computing $\mathbf{k}_1$

$$\mathbf{k}_1 = \mathbf{f}(t_0, \mathbf{y}_0) = \begin{bmatrix} -2(1) \\ -5(1) \end{bmatrix} = \boxed{\begin{bmatrix} -2 \\ -5 \end{bmatrix}}$$

Step 2: Computing $\mathbf{k}_2$

Formula: $\mathbf{k}_2 = \mathbf{f}\left(t_0 + \frac{h}{2}, \mathbf{y}_0 + \frac{h}{2}\mathbf{k}_1\right)$

$$\frac{h}{2}\mathbf{k}_1 = 0.05 \begin{bmatrix} -2 \\ -5 \end{bmatrix} = \begin{bmatrix} -0.1 \\ -0.25 \end{bmatrix}$$

$$\mathbf{y}_0 + \frac{h}{2}\mathbf{k}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} -0.1 \\ -0.25 \end{bmatrix} = \begin{bmatrix} 0.9 \\ 0.75 \end{bmatrix}$$

$$\mathbf{k}_2 = \mathbf{f}\left(t_0 + 0.05, \begin{bmatrix} 0.9 \\ 0.75 \end{bmatrix}\right) = \begin{bmatrix} -2(0.9) \\ -5(0.75) \end{bmatrix} = \boxed{\begin{bmatrix} -1.8 \\ -3.75 \end{bmatrix}}$$

RK4 Derivation: Computing $\mathbf{k}_3$ and $\mathbf{k}_4$

Step 3: Computing $\mathbf{k}_3$

Formula: $\mathbf{k}_3 = \mathbf{f}\left(t_0 + \frac{h}{2}, \mathbf{y}_0 + \frac{h}{2}\mathbf{k}_2\right)$

$$\frac{h}{2}\mathbf{k}_2 = 0.05 \begin{bmatrix} -1.8 \\ -3.75 \end{bmatrix} = \begin{bmatrix} -0.09 \\ -0.1875 \end{bmatrix}$$

$$\mathbf{y}_0 + \frac{h}{2}\mathbf{k}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} -0.09 \\ -0.1875 \end{bmatrix} = \begin{bmatrix} 0.91 \\ 0.8125 \end{bmatrix}$$

Step 4: Computing $\mathbf{k}_4$

Formula: $\mathbf{k}_4 = \mathbf{f}(t_0 + h, \mathbf{y}_0 + h\mathbf{k}_3)$

$$\mathbf{k}_3 = \begin{bmatrix} -2(0.91) \\ -5(0.8125) \end{bmatrix} = \begin{bmatrix} -1.82 \\ -4.0625 \end{bmatrix}$$

$$h\mathbf{k}_3 = 0.1 \begin{bmatrix} -1.82 \\ -4.0625 \end{bmatrix} = \begin{bmatrix} -0.182 \\ -0.40625 \end{bmatrix}$$

$$\mathbf{y}_0 + h\mathbf{k}_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} -0.182 \\ -0.40625 \end{bmatrix} = \begin{bmatrix} 0.818 \\ 0.59375 \end{bmatrix}$$

$$\mathbf{k}_4 = \begin{bmatrix} -2(0.818) \\ -5(0.59375) \end{bmatrix} = \begin{bmatrix} -1.636 \\ -2.96875 \end{bmatrix}$$

RK4 Derivation: Combining Slopes

Step 5: Computing final weighted sum for y_1

Formula: $y_1 = y_0 + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$

$$\begin{aligned}\sum \mathbf{k} &= \begin{bmatrix} -2 \\ -5 \end{bmatrix} + 2 \begin{bmatrix} -1.8 \\ -3.75 \end{bmatrix} + 2 \begin{bmatrix} -1.82 \\ -4.0625 \end{bmatrix} + \begin{bmatrix} -1.636 \\ -2.96875 \end{bmatrix} \\ &= \begin{bmatrix} -2 \\ -5 \end{bmatrix} + \begin{bmatrix} -3.6 \\ -7.5 \end{bmatrix} + \begin{bmatrix} -3.64 \\ -8.125 \end{bmatrix} + \begin{bmatrix} -1.636 \\ -2.96875 \end{bmatrix} = \begin{bmatrix} -10.876 \\ -23.59375 \end{bmatrix}\end{aligned}$$

$$\begin{aligned}\mathbf{y}_1 &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \frac{0.1}{6} \begin{bmatrix} -10.876 \\ -23.59375 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} -0.18127 \\ -0.39323 \end{bmatrix} = \boxed{\begin{bmatrix} 0.81873 \\ 0.60677 \end{bmatrix}}\end{aligned}$$

Accuracy Check at $t = 0.1$

Exact Solutions: $y_1(0.1) = e^{-0.2} \approx 0.81873$, $y_2(0.1) = e^{-0.5} \approx 0.60653$

RK4 Estimate: $y_1 \approx 0.81873$, $y_2 \approx 0.60677$

RK4 is highly accurate after just one step!

RK4 Stability

To study stability, we use the standard linear test equation:

$$y' = \lambda y \implies f(t, y) = \lambda y$$

The 4th-order Runge-Kutta formulas are:

$$k_1 = f(t_n, y_n)$$

$$k_2 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right)$$

$$k_3 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_2\right)$$

$$k_4 = f(t_n + h, y_n + hk_3)$$

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

RK4 Stability: Computing k_1 and k_2

Substitute $f(t, y) = \lambda y$ into the RK4 slopes:

Step 1: Computing k_1

$$k_1 = \lambda y_n$$

Step 2: Computing k_2

We substitute k_1 :

$$\begin{aligned} k_2 &= \lambda \left(y_n + \frac{h}{2} k_1 \right) \\ &= \lambda \left(y_n + \frac{h}{2} \lambda y_n \right) \\ &= \lambda y_n \left(1 + \frac{h\lambda}{2} \right) \\ \mathbf{k}_2 &= \lambda \mathbf{y}_n \left(\mathbf{1} + \frac{\mathbf{z}}{2} \right) \end{aligned}$$

Define $z = h\lambda$:

RK4 Stability: Computing k_3

Step 3: Computing k_3

Substituting k_2 and use $\frac{h\lambda}{2} = \frac{z}{2}$:

$$\begin{aligned}k_3 &= \lambda \left(y_n + \frac{h}{2} k_2 \right) \\&= \lambda \left[y_n + \frac{h}{2} \lambda y_n \left(1 + \frac{z}{2} \right) \right] \\&= \lambda y_n \left[1 + \frac{z}{2} \left(1 + \frac{z}{2} \right) \right]\end{aligned}$$

Expanding:

$$\mathbf{k}_3 = \lambda \mathbf{y}_n \left(\mathbf{1} + \frac{\mathbf{z}}{2} + \frac{\mathbf{z}^2}{4} \right)$$

RK4 Stability: Computing k_4

Step 4: Computing k_4

Substituting k_3 and use $h\lambda = z$:

$$\begin{aligned}k_4 &= \lambda(y_n + hk_3) \\&= \lambda \left[y_n + h\lambda y_n \left(1 + \frac{z}{2} + \frac{z^2}{4} \right) \right] \\&= \lambda y_n \left[1 + z \left(1 + \frac{z}{2} + \frac{z^2}{4} \right) \right]\end{aligned}$$

Expanding:

$$\mathbf{k}_4 = \lambda \mathbf{y}_n \left(\mathbf{1} + \mathbf{z} + \frac{\mathbf{z}^2}{2} + \frac{\mathbf{z}^3}{4} \right)$$

RK4 Stability: Final Substitution

Step 5: Substituting into RK4 update

Recall $y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$. Substitute the k 's:

$$\begin{aligned}y_{n+1} &= y_n + \frac{h}{6}\lambda y_n \left[1 + 2\left(1 + \frac{z}{2}\right) + 2\left(1 + \frac{z}{2} + \frac{z^2}{4}\right) + \left(1 + z + \frac{z^2}{2} + \frac{z^3}{4}\right) \right] \\&= y_n + \frac{z}{6}y_n \left[1 + (2 + z) + \left(2 + z + \frac{z^2}{2}\right) + \left(1 + z + \frac{z^2}{2} + \frac{z^3}{4}\right) \right]\end{aligned}$$

Combine like terms:

- **Constants:** $1 + 2 + 2 + 1 = 6$
- **z -terms:** $z + z + z = 3z$
- **z^2 -terms:** $\frac{z^2}{2} + \frac{z^2}{2} = z^2$
- **z^3 -term:** $\frac{z^3}{4}$

$$y_{n+1} = y_n + y_n \frac{z}{6} \left(6 + 3z + z^2 + \frac{z^3}{4} \right)$$

RK4 Stability: The Multiplier

Step 6: Further distributing $\frac{z}{6}$

$$\begin{aligned}y_{n+1} &= y_n + y_n \left[\frac{z}{6}(6) + \frac{z}{6}(3z) + \frac{z}{6}(z^2) + \frac{z}{6} \left(\frac{z^3}{4} \right) \right] \\&= y_n + y_n \left(z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24} \right)\end{aligned}$$

Factor out y_n :

$$y_{n+1} = y_n \left(1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24} \right)$$

So, $y_{n+1} = R(z)y_n$, where the RK4 stability multiplier is:

RK4 Stability Function

$$R(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}$$

RK4 Stability Condition

For the numerical solution to decay and remain stable, the multiplier must have an absolute value strictly less than 1:

$$|R(z)| < 1$$

For our two-dimensional system:

$$y_1' = -2y_1 \implies \lambda_1 = -2$$

$$y_2' = -5y_2 \implies \lambda_2 = -5$$

Since $z = h\lambda$, we have:

$$z_1 = -2h$$

$$z_2 = -5h$$

So, RK4 is stable for the system when **both** conditions are true:

$$|R(-2h)| < 1$$

and

$$|R(-5h)| < 1$$

RK4 Example with $h = 0.5$

Earlier, we saw that the Euler method was unstable at $h = 0.5$. Let's check if RK4 is stable or not:

RK4 stability condition:

$$R(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24} \quad \text{where} \quad |R(z)| < 1$$

For our 2D system ($y'_1 = -2y_1$, $y'_2 = -5y_2$), we have:

$$\lambda_1 = -2, \quad \lambda_2 = -5$$

With $h = 0.5$ and $z = h\lambda$, we evaluate:

$$z_1 = 0.5(-2) = -1$$

$$z_2 = 0.5(-5) = -2.5$$

We must check both $|R(-1)| < 1$ and $|R(-2.5)| < 1$.

RK4 Example with $h = 0.5$: Compute $R(-1)$

Substituting $z = -1$ into the stability multiplier:

$$\begin{aligned} R(-1) &= 1 + (-1) + \frac{(-1)^2}{2} + \frac{(-1)^3}{6} + \frac{(-1)^4}{24} \\ &= 1 - 1 + \frac{1}{2} - \frac{1}{6} + \frac{1}{24} \\ &= \frac{1}{2} - \frac{1}{6} + \frac{1}{24} \end{aligned}$$

LCM:

$$\begin{aligned} R(-1) &= \frac{12}{24} - \frac{4}{24} + \frac{1}{24} \\ &= \frac{9}{24} = \frac{3}{8} \end{aligned}$$

Component 1 Stability

$$\boxed{R(-1) = 0.375} \implies |0.375| < 1$$

The first component is **stable**.

RK4 Example with $h = 0.5$: Compute $R(-2.5)$

Substituting $z = -2.5$ into the stability multiplier:

$$R(-2.5) = 1 + (-2.5) + \frac{(-2.5)^2}{2} + \frac{(-2.5)^3}{6} + \frac{(-2.5)^4}{24}$$

Calculation:

- $(-2.5)^2 = 6.25 \implies \frac{6.25}{2} = 3.125$
- $(-2.5)^3 = -15.625 \implies \frac{-15.625}{6} \approx -2.6042$
- $(-2.5)^4 = 39.0625 \implies \frac{39.0625}{24} \approx 1.6276$

Substituting back into the equation:

$$R(-2.5) \approx 1 - 2.5 + 3.125 - 2.6042 + 1.6276$$

RK4 Example with $h = 0.5$: Final Conclusion

Combine the terms step by step for $R(-2.5)$:

$$1 - 2.5 = -1.5$$

$$-1.5 + 3.125 = 1.625$$

$$1.625 - 2.6042 = -0.9792$$

$$-0.9792 + 1.6276 = 0.6484$$

Component 2 Stability

$$\boxed{R(-2.5) \approx 0.6484} \implies |0.6484| < 1$$

The second component is also **stable**.

Comparison with Euler

Both components are stable under RK4 at this step size.

$$\boxed{\text{Euler is unstable at } h = 0.5} \text{ vs. } \boxed{\text{RK4 is stable at } h = 0.5}$$

Extension to a 3D System

Let us consider a 3-dimensional system:

$$\begin{cases} y_1' = -2y_1 \\ y_2' = -5y_2 \\ y_3' = -8y_3 \end{cases}$$

For the **Euler stability**:

$$|1 - 2h| < 1 \implies 0 < h < 1$$

$$|1 - 5h| < 1 \implies 0 < h < 0.4$$

$$|1 - 8h| < 1 \implies 0 < h < 0.25$$

For the entire 3D system to be stable, the step size h must satisfy the strictest condition:

$$0 < h < 0.25$$

Main Point

Adding a faster-decaying component can force a smaller stable step size.

- **Summary:** Euler's method is simple but it can become unstable when the step size is too big, and so fourth-order Runge Kutta uses four slope estimates and can remain stable for a larger step size. So, In a **multidimensional system**, every component must satisfy the stability condition.

Thank you!